NexaConsult Ltd

Network Infrastructure Project

1. Project Overview

This project simulates the internal network infrastructure of **NexaConsult Ltd**, a mid-sized IT consultancy firm. The goal was to design and build a fully functional, segmented, and secure network from scratch using Cisco Packet Tracer — covering everything from physical topology to VLAN configuration, dynamic IP assignment, inter-department routing, access control, and secure remote management.

The project was built step by step, troubleshooting real issues along the way, which made it a genuine learning experience rather than just following a guide.

2. Project Objectives

- Design a realistic multi-department network topology
- Segment departments using VLANs for traffic isolation and security
- Configure Router-on-a-Stick for inter-VLAN routing
- Deploy a centralised DHCP server to automate IP assignment across all VLANs
- Set up DNS and an internal HTTP intranet server
- Implement Access Control Lists (ACLs) to restrict department access
- Enable SSH on all devices and disable Telnet for secure management
- Apply port security on all access switch ports

3. Network Topology

3.1 Devices Used

Device	Model	Role
NexaConsult-R1	ISR 4331	Main company router — handles inter-VLAN routing and NAT
ISP-Router (ISPR)	ISR 4331	Simulated ISP router for WAN connectivity
NexaConsult-Core	3560-24PS	Core multilayer switch — central distribution point
NexaConsult-SW1	2960-24TT	Access switch — Management department

NexaConsult-SW2	2960-24TT	Access switch — IT and Consulting departments
NexaConsult-SW3	2960-24TT	Access switch — HR & Finance and Guest WiFi
NexaConsult-SWS	2960-24TT	Server switch — connects all three servers
Server0	Server-PT	DHCP Server
Server1	Server-PT	DNS Server
Server2	Server-PT	HTTP Intranet Server
Management PC	PC-PT	Test endpoint — VLAN 10
IT PC	PC-PT	Test endpoint — VLAN 20
Consulting PC	PC-PT	Test endpoint — VLAN 30
HR&Finance PC	PC-PT	Test endpoint — VLAN 40
Guest WiFi PC	PC-PT	Test endpoint — VLAN 50

4. IP Addressing Plan

4.1 VLAN Subnets

VLAN	Department	Network	Gateway	DHCP Range
10	Management	192.168.10.0/24	192.168.10.1	192.168.10.10 - .50
20	IT / Admin	192.168.20.0/24	192.168.20.1	192.168.20.10 - .50
30	Consultants	192.168.30.0/24	192.168.30.1	192.168.30.10 - .80
40	HR & Finance	192.168.40.0/24	192.168.40.1	192.168.40.10 - .50
50	Guest WiFi	192.168.50.0/24	192.168.50.1	192.168.50.10 - .80
99	Servers	192.168.99.0/24	192.168.99.1	Static only

4.2 Server Static Addresses

Server	IP Address	Role
Server0	192.168.99.10	DHCP Server
Server1	192.168.99.11	DNS Server
Server2	192.168.99.12	HTTP Intranet Server

5. Configuration Summary

5.1 Device Hardening

All network devices (both routers and all five switches) were configured with consistent security settings including encrypted enable passwords, console passwords, VTY line passwords, service password-encryption, and a MOTD banner warning unauthorised users.

5.2 VLAN Configuration

Six VLANs were created across all switches. The Core switch trunks were configured using 802.1Q encapsulation (required explicitly on the 3560 multilayer switch). Access ports on each switch were assigned to the correct VLAN for each department.

5.3 Router-on-a-Stick

Instead of using separate physical interfaces for each VLAN, subinterfaces were created on G0/0/1 of NexaConsult-R1 — one per VLAN (.10 through .99). Each subinterface was assigned the gateway IP for its respective VLAN, enabling traffic to route between departments through a single physical link.

5.4 DHCP

A centralised DHCP server (Server0, 192.168.99.10) was configured with separate address pools for each VLAN. IP helper addresses were added to each router subinterface to forward DHCP broadcast requests from PCs across to the server, since broadcasts cannot cross routers by default.

5.5 DNS and HTTP

Server1 was configured as the DNS server with A records pointing nexaconsult.local and www.nexaconsult.local to the HTTP server at 192.168.99.12. Server2 was configured as the HTTP server hosting a custom NexaConsult intranet homepage.

5.6 Access Control Lists (ACLs)

Two extended named ACLs were configured on the router to enforce department access restrictions:

- BLOCK-GUEST: Applied inbound on G0/0/1.50 — prevents Guest WiFi from reaching any internal department or server VLAN
- BLOCK-HR: Applied inbound on G0/0/1.40 — prevents HR & Finance from communicating with other departments while still allowing server access

5.7 SSH

SSH v2 was enabled on all devices. RSA keys were generated using a 1024-bit key size with the domain name nexaconsult.local. VTY lines were restricted to SSH only (transport input ssh), Telnet was disabled, and a local admin account was created for authentication.

5.8 Port Security

Port security was applied to all access switch ports connecting to PCs. Each port was configured to allow a maximum of one MAC address, using sticky learning to automatically remember the connected device. Violation mode was set to shutdown, meaning the port automatically disables itself if an unauthorised device is plugged in.

6. Problems Encountered

Problem 1 — Trunk encapsulation error on the Core switch

Error: "An interface whose trunk encapsulation is Auto cannot be configured to trunk mode"

When setting trunk ports on the 3560 multilayer switch, the command switchport mode trunk was rejected. Unlike the 2960 access switches which only support 802.1Q and set it automatically, the 3560 supports multiple encapsulation types and requires you to specify it manually first. The fix was to run switchport trunk encapsulation dot1q before setting the mode to trunk.

Problem 2 — DHCP failing with APIPA address (169.254.x.x)

After configuring the DHCP server and helper addresses, the Management PC kept getting a 169.254.x.x address instead of a proper 192.168.10.x address. This was a multi-step troubleshooting process. The root cause turned out to be that the trunk port on the Core switch connecting to Router1 was configured on G0/1, but the actual cable was physically connected to Fa0/5. Running show cdp neighbors on the Core revealed the real port, and once the trunk was configured on the correct interface (Fa0/5), DHCP started working.

Problem 3 — Wrong device for ACL configuration

When trying to fix the HR & Finance ACL, the commands were accidentally entered on NexaConsult-SW3 (a switch) instead of NexaConsult-R1 (the router). Switches cannot have subinterfaces like g0/0/1.40, so the interface command failed. No damage was done since switches simply threw errors and ignored the commands. The fix was to close the wrong CLI window and open the correct router.

Problem 4 — ACL name case mismatch

The BLOCK-HR ACL was created with the name block-hr (lowercase) but applied to the interface as BLOCK-HR (uppercase). Cisco IOS treats ACL names as case-sensitive, so the ACL existed but was not actually being applied to the traffic. The fix was to delete the lowercase version and recreate it with the correct uppercase name, then reapply it to the interface.

Problem 5 — Router hostname showing incorrectly on canvas

After configuring the hostname NexaConsult-R1 in the CLI, the canvas label showed Router3 instead. This was a Packet Tracer display quirk — the CLI hostname was set correctly and visible in all show commands, but the canvas label did not update automatically. This was a cosmetic issue only and did not affect functionality.

Problem 6 — HTTP server not loading

The HTTP intranet page on Server2 did not load when browsed to from a PC. This appeared to be related to Server2 not having its static IP properly configured at the time of testing. The DNS and DHCP configuration were both verified as correct. This item was noted for follow-up but did not affect the core network functionality tests.

7. Test Results

Test	Description	Expected Result	Actual Result
Test 1	Inter-VLAN routing — Management to Consulting	Ping succeeds	PASSED (3/4 packets, first lost to ARP)
Test 2	ACL — Guest blocked from Management	Ping fails	PASSED (100% packet loss)
Test 3	ACL — HR blocked from Consultants	Ping fails	PASSED (100% packet loss)
Test 4	DHCP on all 5 PCs	Each PC gets correct subnet IP	PASSED
Test 5	HTTP intranet browsing	Page loads in browser	INCOMPLETE

8. What I Learned

- VLANs are not just a configuration step — they are the foundation of a secure network. Without them, every device on the network can see every other device's traffic.
- Router-on-a-stick is a cost-effective way to route between VLANs without needing multiple physical interfaces — one cable, multiple subinterfaces.
- DHCP broadcasts do not cross routers by default. The ip helper-address command is what bridges that gap and makes centralised DHCP possible across a multi-VLAN network.
- show cdp neighbors is an incredibly useful diagnostic command — it tells you exactly which device is connected to which port, which helped solve the trunk port issue.
- ACL names are case-sensitive in Cisco IOS. A mismatch between the name you create and the name you apply results in a silent failure — the ACL exists but does nothing.
- Troubleshooting a real network issue (like DHCP failure) requires a systematic approach — check each layer one at a time rather than randomly trying fixes.
- SSH setup requires a domain name and RSA key generation before it will work. Without the domain name, the crypto key generate rsa command fails.

9. Conclusion

The NexaConsult Ltd network project successfully demonstrates a realistic enterprise-level network simulation covering the core topics of the CCNA curriculum. The network is fully segmented by department, automatically assigns IP addresses, enforces access restrictions between departments, and is securely managed via SSH with Telnet disabled on all devices.

The troubleshooting challenges encountered throughout the project — particularly the DHCP failure caused by a trunk port on the wrong interface — were the most valuable part of the experience. Real network issues rarely come with obvious error messages, and learning to use diagnostic commands like show cdp neighbors, show interfaces trunk, and show vlan brief to methodically isolate problems is a skill that goes well beyond any single configuration step.

The one outstanding item is the HTTP server, which will be revisited to ensure Server2 has its static IP correctly bound and the intranet page is accessible from all internal VLANs.